



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, NAGPUR
Department of Computer Science & Engineering
CSL 301: Database Management Systems

Mid-Sem Exam

Duration: 1:30 hour

Max. Marks:

Max Marks: 30

Semester – V

Date: 22 Sept 2026 (Tue)

Important Instructions:

- This is a closed book, closed notes examination.
- This question paper comprises a total of 4 questions.
- All the questions are compulsory.

Q1) A university has several departments. Each department offers multiple academic programs such as B.Tech, M.Tech, Ph.D., and certificate courses. Each program consists of several courses. A course may be compulsory for one program but elective for another program. Some courses have prerequisites, and a course may have multiple prerequisite courses. Students are admitted into a specific program and department. A student may enroll in several courses in a semester. Each course is taught by one or more faculty members. A faculty member may teach multiple courses in different semesters. For every course enrollment, the system stores attendance percentage, internal marks, end-semester marks, grade, and result status. Each department has one Head of Department, who must be a faculty member of that department. A faculty member may also act as a program coordinator for one or more programs. Each course has one course coordinator, but the course coordinator must be one of the faculty members teaching that course. The university also maintains information about classrooms, labs, time slots, and semester-wise time table. A classroom can be assigned to many courses, but no two courses can be scheduled in the same classroom at the same time slot. A faculty member also cannot teach two courses at the same time. (6X2=12M)
CO1

I. Construct the ER diagram for the scenario described above. Classify each entity set explicitly (e.g., strong, weak, etc). Annotate all relationships with their cardinality ratios (1:1, 1:N, M: N) and participation constraints (total vs. partial). Clearly indicate primary keys, and identify any composite, multivalued, or derived attributes as appropriate.

II. From your ER model, derive the minimum set of relations required to represent the system without loss of information. For each relation, present the schema in the form RelationName(attributes), underlining the primary key and listing all foreign keys together with their referenced relation(s) attribute(s) (i.e., show the linking information).

Q2) Analyze the three-schema architecture of a DBMS by elaborating the roles of the external, conceptual, and internal levels. Examine how the mappings between these levels enable logical and physical data independence, using suitable examples. (3M)
CO1

Q3) Consider the following relational schema for a Library Management System: (1+2+2+2+2=9M)
STUDENT(StudentID, StudentName, *DepartmentID*, Semester, CGPA) CO2

DEPARTMENT(DepartmentID, DepartmentName)

FACULTY(FacultyID, FacultyName, *DepartmentID*, Designation)

COURSE(CourseID, CourseName, Credits, *DepartmentID*)

TEACHES(FacultyID, CourseID, AcademicYear, Semester)

ENROLLMENT(StudentID, CourseID, AcademicYear, Semester, Grade)

In the relational schema above, the primary keys are shown in **bold and underlined**. The foreign keys are represented in *italic* form.

Write SQL and RA (Relational Algebra) for the following:

- I. Display the names of all courses along with the department that offers them.
- II. Find students who have not enrolled in any course.
- III. Find students who are enrolled in at least one course offered by a department different from their own department.
- IV. Find students whose CGPA is greater than the average CGPA of all students.
- V. Find students who have enrolled in all courses offered by the CSE department.

Q4) Consider relation R(A, B, C, D, E, F, G, H) with functional dependencies: (2+2+1+1=6M)
 $F = \{ A \rightarrow BC, B \rightarrow DC, CD \rightarrow AF, E \rightarrow CF, F \rightarrow B, G \rightarrow H, H \rightarrow A \}$ CO2

Given a relation (R) and a set of functional dependencies (F):

- I. Compute the minimal cover of (F), showing all intermediate steps. Identify which functional dependencies are preserved and which are lost after decomposition.
- II. Determine all candidate keys of (R).
- III. Decompose (R) into BCNF using the minimal cover obtained in Part I.
- IV. Analyze the given BCNF decomposition and determine whether it satisfies the lossless-join property. Justify your answer